

Fig. 14: Simulation window

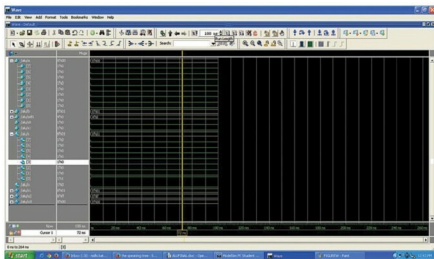


Fig. 15: Wave window

Simulating the ALU design

1. Click on Library menu from the main window and then click on the plus (+) sign next to the work library. You should see the name Top_ALU code that we have just

compiled (Fig. 10).

2. Double-click on alu to load the file. This should open a third tab sim in the main window.

3. Go to Add -> Wave-> All items in region options (Fig. 11).

EFY Note

The source codes of this project are included in this month's EFY DVD and are also available for free download at source.efymag.com

4. Select the signals that you want to monitor for simulation purposes. Select these as shown in Fig. 12.

5. Provide values manually to monitor the simulation of the eight-bit ALU design.

Right-click on the selected signals and click on Force (Fig. 13).

After providing values to selected signals, we are now ready to simulate our design by clicking Run in the simulation window as shown in Fig. 14.

Now, click on Zoom Full from the wave window. Your simulation output waveform will be as shown in Fig. 15.

Thus, the result of the ALU design is verified from this output waveform.

Future applications

The ALU described here can be further refined to make more advanced-level projects with some more complex logics in order to handle complex calculations in the CPUs. ●



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